



# SUJET STATAPP 305 - 2026-2027

**Sujet proposé par :** Rufat Samuel - [samuel.rufat@polytechnique.edu](mailto:samuel.rufat@polytechnique.edu)  
**Organisme :** CREST  
**Catégorie :** Environnement, Energie, Mobilité  
**Sujet :**  
**Disaster Risk Reduction and Climate Change Adaptation in European Metropolises**

## Description du sujet :

Despite significant investments in Disaster Risk Reduction and Climate Change Adaptation, current policies often fail to reduce losses effectively due to a mismatch between theoretical models and empirical reality. The EU vision of a disaster- and climate-resilient society cannot be achieved with the current approach, as it overlooks actual human behaviours.

The European commission-funded ERC FiBeGa project is hosted at the Center for Research in Economics and Statistics (CREST); it develops spatially explicit, interactive simulation tools for the short and long term, enabling decision-makers to refine their strategies and policies.

This project involves an interdisciplinary team in computational social sciences to generate new empirical evidence on adaptive behaviour focusing on floods and heatwaves in large European metropolises, detect the local predictors with reproducible data analytics to refine existing theories, and develop spatially-explicit simulation tools to reduce future losses.

## Données du sujet :

The project combines a wide range of data:

- ? Survey data, large samples (n=5,000) from several European metropolises;
- ? Census data (grid cell population and housing results);
- ? Healthcare and hospitalisation data;
- ? Climate and disasters impact data.

## Modèles envisagés du sujet :

Proficiency with R studio or Python and standard office software is expected. Experience with spatially-explicit agent-based modelling (ABM) can be acquired.

## Bibliographie du sujet :

- Jézéquel, A., Bastos, A., Wilson, A. M., Rufat, S., et al. (2024). Broadening the scope of anthropogenic influence in extreme event attribution. *Environmental Research: Climate*, 10.1088/2752-5295/ad7527
- Rufat, S., Hudson, P., & Enderlin, E. (2025). Theoretical frameworks of risk perception and protective behaviour: an empirical comparison. *Natural Hazards*, 121, 14697-14767.
- Rufat, S., Metzger, P. (2024). *Vulnerability, Territory, Population: From Critique to Public Policy*, Wiley.
- Rufat, S., Robinson, P. J., & Botzen, W. J. (2024). Insights into the complementarity of natural disaster insurance purchases and risk reduction behavior. *Risk Analysis*, 44(1), 141-154.
- Rufat, S., & Howe, P. D. (2023). Small-area estimations from survey data for high-resolution maps of urban flood risk perception and adaptive behavior. *Annals of the American Association of Geographers*, 113(2), 425-447.
- Rufat, S., & Botzen, W. W. (2022). Drivers and dimensions of flood risk perceptions: Revealing an implicit selection bias and lessons for policies. *Global Environmental Change*, 73, 102465.

**Notation du sujet :**

**Commentaires :**